

FINAL 45-MINUTE PRESENTATION SCRIPT

“Attention Is All You Need” (Vaswani et al., 2017)

Complete Paper Explanation — From Authors to End

Explained for a Cancer Research Centre & Hospital Team

HOW TO USE THIS

- This is a **complete spoken presentation** you can deliver end-to-end
 - You may read, paraphrase, or convert each section into slides
 - Designed for **45 minutes total**
 - Audience: clinicians, researchers, data/AI teams, leadership
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0–4 MIN | TITLE, AUTHORS & CONTEXT

Slide 1: Title

Attention Is All You Need Vaswani et al., NeurIPS 2017

What I say:

Today I will explain a landmark AI research paper called “*Attention Is All You Need*”, published in 2017 by researchers from Google Brain and Google Research. The main authors include Ashish Vaswani, Noam Shazeer, Jakob Uszkoreit, and their team.

This paper introduced the **Transformer architecture**, which later became the foundation for ChatGPT, modern medical AI systems, and large-scale clinical intelligence platforms.

Why authors matter:

- Google Brain → real-world, large-scale AI
- Focus on **efficiency, scalability, and reliability**

Cancer-centre relevance:

Almost every modern AI system used in radiology, pathology, genomics, and clinical text today is based on this paper.



4–8 MIN | WHY THIS PAPER WAS NEEDED (INTRODUCTION)

Slide 2: AI Before Transformers

What the paper explains:

- Dominant models: RNNs / LSTMs
- Sequential processing
- Slow training
- Difficulty with long sequences

What I say:

Before 2017, AI models processed information step by step, like reading a patient file one word at a time. As data became longer — clinical notes, genomic sequences, imaging studies — these models struggled.

Cancer-hospital problem:

- Long patient histories
- Multi-year follow-up data
- Whole-slide pathology images

The paper asks a simple but powerful question: *Do we really need recurrence at all?*



8–13 MIN | THE CORE IDEA: ATTENTION

Slide 3: What Is Attention?

Paper concept:

Attention = focusing on what matters most

What I say:

Attention allows the model to look at the entire input at once and decide which parts are most relevant. Instead of remembering everything equally, it prioritizes.

Clinical analogy:

- Radiologist focuses on suspicious lesions
- Pathologist focuses on abnormal tissue regions
- Oncologist focuses on high-risk factors

Cancer research value:

This mirrors how clinicians and researchers think.



13–18 MIN | SELF-ATTENTION (KEY CONTRIBUTION)

Slide 4: Self-Attention Explained

What the paper introduces:

- Each element attends to every other element

What I say:

In self-attention, every part of the data asks every other part: “How relevant are you to me?”
This allows the model to understand global relationships.

Cancer examples:

- Relationships between lesions in a scan
- Gene–gene interactions in genomics
- Context across sentences in clinical notes

Why it matters:

Cancer is a disease of interactions, not isolated signals.



18–22 MIN | ATTENTION EQUATION (CONCEPTUAL)

Slide 5: Scaled Dot-Product Attention

Paper equation (do not derive):

$\text{Attention}(Q, K, V)$

What I say:

Queries ask questions, keys represent context, and values carry information. The model scores relevance, normalizes it, and combines information intelligently.

Clinical meaning:

- Mimics diagnostic prioritization
- Enables explainability via attention maps

Hospital impact:

- Trustworthy AI
 - Easier validation and audit
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 22-26 MIN | MULTI-HEAD ATTENTION**Slide 6: Why Multiple Heads?****Paper insight:**

- One attention view is insufficient

What I say:

Multi-head attention allows the model to look at the same data from multiple perspectives at the same time.

Tumor board analogy:

- One head: radiology
- One head: pathology
- One head: genomics
- One head: clinical history

The final decision integrates all perspectives.

 26-31 MIN | TRANSFORMER ARCHITECTURE**Slide 7: Encoder-Decoder Structure****Paper design:**

- Encoder stack: understanding
- Decoder stack: generation

What I say:

The encoder converts raw input into a deep understanding. The decoder uses that understanding to generate outputs step by step.

Hospital mapping:

- Encoder: patient data ingestion (images, text, omics)
 - Decoder: reports, summaries, predictions
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31–34 MIN | POSITIONAL ENCODING

Slide 8: Understanding Order Without Recurrence

Problem:

- Attention has no sense of order

Solution:

- Positional encoding

What I say:

Positional encoding allows the model to understand sequence order — critical for disease progression, treatment timelines, and follow-up data.



34–38 MIN | TRAINING DETAILS → CLINICAL MEANING

Slide 9: Training Choices in the Paper

Paper includes:

- Adam optimizer
- Learning-rate warm-up
- Large batch sizes
- Efficient GPU usage

What I say:

These are not just technical tricks. They ensure stable learning, efficiency, and scalability.

Cancer-centre impact:

- Reliable AI behavior
 - Lower infrastructure cost
 - Faster experimentation
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38–41 MIN | RESULTS & EVALUATION

Slide 10: Paper Results

What the paper shows:

- State-of-the-art translation performance
- Faster training than RNNs

Healthcare translation:

- Architecture generalizes well
 - Suitable for complex medical data
 - Proven scalability
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41–44 MIN | WHY THIS PAPER MATTERS FOR CANCER CENTRES

Slide 11: Strategic Importance

What I say:

This paper is the foundation of modern medical AI. It enables multimodal, scalable, explainable systems — exactly what cancer care and research demand.

Applications:

- Radiology AI
 - Digital pathology
 - Genomics & biomarkers
 - Clinical trial matching
 - MDT decision support
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44–45 MIN | FINAL CONCLUSION

Slide 12: Take-Home Message

Closing statement (memorize):

Attention Is All You Need showed that attention alone is sufficient to build fast, scalable, and interpretable AI systems — and this idea now powers the future of cancer research and clinical intelligence.

END OF COMPLETE 45-MINUTE PRESENTATION